

# OE-EC604C — Object Oriented Programming

Semester VI • 3L:0T:0P • 3 Credits

## Module 1 Exam Notes: Programming Paradigm & OOP Concepts

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## 1. Evolution of Programming Paradigm

A programming paradigm defines the style, approach, and philosophy of designing and writing software code. As programs grew in size and complexity, paradigms evolved to improve code readability, reuse, and maintainability.

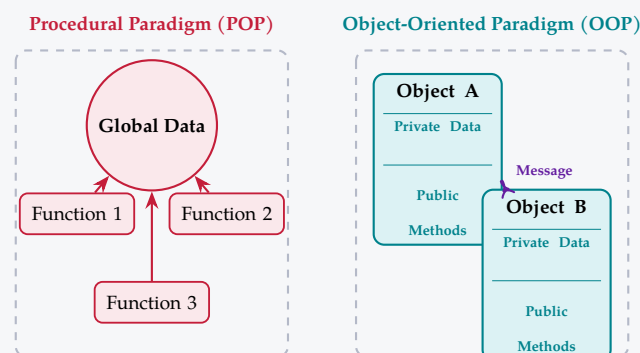
### 1.1 Paradigms Evolution Timeline

- 1. Unstructured Programming:** The earliest style where the program consists of a single continuous block of statements (often containing global variables) using jump statements like `goto` for flow control. It becomes difficult to maintain as code grows ("spaghetti code").
- 2. Procedural / Structured Programming:** Programs are organized into subroutines, modules, or functions. Flow control is structured using blocks like loops (`for`, `while`) and conditional statements (`if`, `switch`). Focus is on the **process or algorithm** rather than data.
- 3. Object-Oriented Programming (OOP):** A paradigm where programs are organized around **objects** (representing real-world entities) containing data (attributes) and methods (behaviors). Focus is on **data security and encapsulation** rather than procedures.

### 1.2 Structured vs. Object-Oriented Development

- **Structured/Procedure-Oriented Programming (POP):** Views a program as a sequence of procedural steps or function calls. Global data is vulnerable because it can be accessed and modified freely by any function.
- **Object-Oriented Programming (OOP):** Views a program as a collection of self-contained, interacting objects. Data is tied closely to the functions that operate on it, preventing accidental outside modification.

#### Procedural Paradigm (POP) vs. Object-Oriented Paradigm (OOP)



## 2. Core Object-Oriented Programming Concepts

OOP seeks to model software systems in terms of real-world entities. Below are the foundational concepts of OOP:

## 2.1 Objects and Classes

**Class:** A user-defined data type that acts as a blueprint, template, or prototype for creating objects. It defines the properties (data members) and actions (member functions) common to all objects of that class.

**Object:** A basic run-time entity of an object-oriented system that represents a real-world entity. It is an instance of a class, possessing its own state (data values) and behavior.

## 2.2 Data Encapsulation and Data Abstraction

**Data Encapsulation:** The mechanism of binding together code (member functions) and the data it manipulates (data members) into a single unit (class). It implements data hiding, keeping data safe from outside interference.

**Data Abstraction:** The act of representing essential features of a system without including the background details or implementation explanations. Users interact only with the interface, hiding complex execution.

## 2.3 Inheritance

**Inheritance:** The process by which objects of one class acquire the properties and behaviors of another class. It supports hierarchical classification and promotes **code reusability** by allowing new classes (derived classes) to extend existing ones (base classes).

**Base Class (Parent Class):** The class whose properties are inherited.

**Derived Class (Child Class):** The class that inherits from the base class.

## 2.4 Polymorphism

**Polymorphism:** The ability of an interface to take multiple forms (from Greek, meaning "many forms"). It allows a single function name, operator symbol, or interface to behave differently depending on the context.

There are two main classifications:

- **Compile-Time (Static) Polymorphism:** Resolved at compile time (e.g., function overloading, operator overloading).
- **Run-Time (Dynamic) Polymorphism:** Resolved at runtime during execution (e.g., virtual functions, method overriding).

## 2.5 Dynamic Binding and Message Passing

**Dynamic Binding:** The mechanism where the code associated with a given function call is not resolved until runtime. It is associated with inheritance and run-time polymorphism.

**Message Passing:** The communication process between objects. Objects interact by sending requests to execute methods of other objects, passing arguments and receiving results.

## Quick Revision — Module I

Paradigm & OOP Concepts | OE-EC604C

### Key Concepts at a Glance

|                             |                                                                                                                                                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Class Structure</b>      | <pre>class ClassName { private: // data public: // methods };</pre>                                                                                                 |
| <b>The 4 Pillars of OOP</b> | <b>Encapsulation</b> (data hiding), <b>Abstraction</b> (interface separation), <b>Inheritance</b> (code reuse), <b>Polymorphism</b> (many forms).                   |
| <b>Access Modifiers</b>     | <b>private:</b> Accessible only within class (default).<br><b>public:</b> Accessible from anywhere.<br><b>protected:</b> Accessible within class and child classes. |

### One-Line Definitions

- **Paradigm:** A structured method or style of programming defining program organization.
- **Class:** A user-defined logical blueprint defining data and member functions.
- **Object:** A physical instance of a class representing a real-world entity.
- **Encapsulation:** Wrapping data and code together into a single unit (class) to prevent global access.
- **Data Hiding:** Restricting direct access to internal data members using access specifiers.
- **Abstraction:** Hiding background implementation details and exposing only the interface.
- **Inheritance:** The mechanism of deriving new classes from existing classes to achieve code reuse.
- **Polymorphism:** The capability of a function, operator, or object to act differently in different contexts.
- **Dynamic Binding:** Resolving method calls at runtime rather than compile time.
- **Message Passing:** Communication model where objects invoke member functions of other objects.

### Frequently Asked / Viva Questions

#### Q1. State the main limitation of Unstructured Programming.

Unstructured programming relies heavily on jump commands (like `goto`), making the control flow complex and highly difficult to trace, maintain, or debug.

#### Q2. What is the core difference in how POP and OOP view data?

**Frequently Asked / Viva Questions (contd.)**

Procedure-Oriented Programming (POP) views data as passive elements manipulated by active procedures. OOP views data as critical assets tied securely to active functions.

**Q3. Define the difference between a Class and an Object.**

A class is a logical template or blueprint (allocates no memory). An object is a physical instance of that template created in memory (allocates memory).

**Q4. Why does OOP offer better data security than POP?**

OOP hides data members inside classes using the `private` access specifier, preventing outside functions from modifying the variables. POP global data is accessible to all.

**Q5. How does Encapsulation differ from Abstraction?**

Encapsulation is the *wrapping* or *containment* of data (data hiding). Abstraction is the *hiding of complexity* by showing only the external interface.

**Q6. What is the primary benefit of Inheritance?**

Code reusability. It allows developer to extend base class properties without rewriting the code, reducing testing and development time.

**Q7. Explain the term "Polymorphism" with a real-world example.**

A single instruction performs different actions based on target. E.g., the "+" symbol performs addition for numbers, but performs concatenation for strings.

**Q8. Define static binding vs. dynamic binding.**

Static binding (early binding) resolves function calls at compile time. Dynamic binding (late binding) resolves function calls at runtime during execution.

**Q9. What is the role of access specifiers in C++?**

Access specifiers (`private`, `public`, `protected`) control the visibility and accessibility of class members from outside classes.

**Q10. What is meant by the term "Message Passing"?**

It is the process by which an object requests execution of a method belonging to another object, passing data parameters and receiving return values.

**Q11. Can you have an object without a class in C++?**

No. In C++, every object must belong to a predefined class which defines its structure and member specifications.

**Q12. What is "Spaghetti Code" and in which paradigm does it commonly occur?**

Spaghetti code is program code with highly convoluted control flow, usually caused by multiple jump statements (`goto`). It occurs in unstructured programming.

## Comparison of POP and OOP Paradigms

| Parameter   | Procedure-Oriented (POP)              | Object-Oriented (OOP)                           |
|-------------|---------------------------------------|-------------------------------------------------|
| Approach    | Top-down design methodology.          | Bottom-up design methodology.                   |
| Focus       | Focuses on the algorithm/process.     | Focuses on the data assets.                     |
| Data Access | Functions share global data (unsafe). | Objects hide data using private scope (secure). |
| Reusability | Limited to function calls.            | Achieved via class inheritance.                 |
| Examples    | C, Pascal, Fortran.                   | C++, Java, Python, C#.                          |